

Food safety in Thailand 5: the effect of washing pesticide residues found in cabbages and tomatoes

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Abstract Plant protection products including pesticides are extensively used in worldwide agriculture. In order to protect consumer health, maximum residue limits (MRL) have been established in many countries and by the Codex Alimentarius Commission. In Thailand, the MRL regulation exists but is not fully enforced. Thus, the purpose of this study was to determine pesticide residues in cabbages and tomatoes sold in Thailand and to evaluate if washing vegetables by running water method can remove pesticide residues. 82 cabbage and 100 tomato samples were collected from markets in Bangkok and the metropolitan Bangkok area. They were extracted and quantitatively analyzed for 42 pesticides by GC-MS/MS. Pesticide residues were detected in all of the cabbage samples, representing a rate of pesticide detection of 100%. These samples contained pesticides at levels that exceeded the recommended MRL, denoting a 100% rate of MRL exceedance. Pesticide residues were also detected in all of tomato samples but incidence of MRL exceedance in tomato samples was 82%. Washing cabbages under running water

significantly reduced carbofuran residues by 93% and dimethoate residues by 65% ($p < 0.05$). In contrast, the running water method did not significantly reduce pesticide residues ($p > 0.05$) in tomato samples. Therefore, this study has documented evidence of remarkable pesticide residues detected in the cabbage and tomato sold in Thailand. These strongly suggest that Thai government authorities should implement the proper routine monitoring program of pesticide residues in vegetables and fruits. Washing vegetables before consumption is advisable as this considerably reduces the risk of exposure to pesticide residues.

Keywords Pesticide residues · Vegetables · Cabbage · Tomato · Food safety

1 Introduction

Nowadays, farmers need pesticides to increase farm productivity by eradicating pests and diseases. In Thailand, pesticides are applied extensively, but not used according to the manufacturer's instructions. Furthermore, 30% of pesticide products marketed in developing countries do not meet accepted quality standards (Pacific PAN 2010). The application of pesticides on vegetables and fruits leaves potentially harmful residues on the produce. Consumption of pesticide contaminated fruits and vegetables is one of the main routes of human exposure to pesticide residues. It is a significant public health risk (Reiler et al. 2015). Since pesticides are widely used in agriculture, there has been growing public awareness and worry about pesticide residues in vegetables and

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fruits and food safety (Jardim et al. 2014; Reiler et al. 2015). The use of pesticides in crops as well as the pesticide residue levels in food are regulated and monitored in developed countries. As a result, low levels of pesticide residues are normally found in food produced in developed countries and are not considered to trigger harmful effects in the consumers (Lozowicka 2015; Nougadère et al. 2012). Pesticide residue monitoring in food commodities is essential to prove if pesticide levels are within the “maximum residue limits (MRL)”. The MRL is the maximum concentration (level) of a pesticide residue, stated as mg/kg, to be legally allowed in food commodities (MacLachlan and Hamilton 2010). MRL values are based on supervised residue experimental data where the pesticide has been applied in compliance with GAP (Good Agricultural Practices). Several developed countries have implemented pesticide monitoring program, and MRL regulation to control pesticide residues in order to ensure compliance with national and international food law (Food and Agriculture Organization 2009; Walorczyk 2008). However, in developing countries such as Thailand, MRL regulations as well as GAP are not fully enforced.

Cabbage (*Brassica oleracea*) and tomato (*Solanum lycopersicum*) are 2 of the most widely consumed vegetables. Cabbage is an economically important crop grown in more than 90 countries around the world (Paramasivam and Banerjee 2013; Wikipedia 2016). In Thailand, cabbages are the usual vegetables of commerce, consumed both as fresh (raw), cooked, pickled and fermented dishes. Cabbage is a decent source of vitamin K, vitamin C and dietary fiber. Tomato is one of the most commonly cultivated vegetables in the world. Tomatoes in several varieties are extensively grown all over the world. The varieties cultivated in Thailand are principally divided into 6 types, based mainly on shape and size. These include beefsteak tomatoes, oxheart tomatoes, plum tomatoes, pear tomatoes, cherry tomatoes and grape tomatoes. In Thai cuisine, tomatoes are used in a broad range of cooked dishes and are also eaten fresh (raw). Nowadays, with western culture influence, many young people especially prefer to eat whole tomatoes in fresh salads (such as cherry and grape tomatoes) and in sandwiches.

Vegetable pesticide residue monitoring programs have been operating in many developed countries (Food and Agriculture Organization 2009; Walorczyk 2008). However, negligible information is available on pesticide residues in developing countries, including Thailand.

Systematic studies are required to confirm the existing status of pesticide contamination in foods, particularly in fruits and vegetables in Thailand (Phopin et al. 2017; Wanwimolruk et al. 2015a, b). The present research study continues the “Food Safety in Thailand” series, aimed at determining pesticide residues in 2 commonly consumed vegetables, cabbages and tomatoes. Both vegetables are important ingredients in a famous Thai dish known as Papaya Salad (“Som-Tum”). This dish is very popular among the Thai people and often eaten as an appetizer. In general, consuming fresh vegetables can potentially be toxic due to pesticide contamination if the vegetables are consumed daily and if they are not washed properly (Hamilton et al. 2004; Hanafi et al. 2016). Thus, the effects of washing cabbages and tomatoes with running water as a method to reduce pesticide residues and to minimize the associated health risk, were also evaluated.

2 Materials and methods

2.1 Chemicals and standards

Sodium chloride and anhydrous magnesium sulphate were purchased from Ajax Finechem, Australia. Primary secondary amine (PSA, particle size 40 µm) and graphite carbon black (GCB) were obtained from Supelco (Sigma-Aldrich Corp., St. Louis, USA). HPLC-grade acetonitrile was purchased from Merck (Darmstadt, Germany). 42 pesticides and two metabolite standards including aldrin, atrazine, captan, carbaryl, carbofuran (and its two metabolites carbofuran-3-hydroxy and carbofuran-3-keto), carbosulfan, chlormefos, chlorpyrifos, chlorothalonil, cyfluthrin, λ -cyhalothrin, cypermethrin, deltamethrin, diazinon, dichlorvos, dicofol, dicrotophos, dimethoate, ethion, fenithion, fenitrothion, fenobucarb, fenvalerate, flumethrin, γ -BHC, heptachlor, isoprocarb, malathion, metalaxyl, methidathion, methomyl, monocrotophos, omethoate, paraoxon-methyl, phosalone, pirimicarb, pirimiphos-methyl and profenofos were purchased from Dr. Ehrenstorfer (Augsburg, Germany). Purity of these pesticide standards was >98%. An individual stock of standard solutions (1000 mg/L) was prepared in acetonitrile. A working surrogate spiking standard solutions were prepared by an appropriate dilution of the stock solutions with acetonitrile. These standard solutions were safe from light and kept frozen at -20°C until analyzed.

2.2 Vegetable samples

A total of 82 cabbage samples and 100 tomato samples were purchased randomly from markets in 7 provinces including Bangkok, Pathumthani, Ayutthaya, Nakhon Pathom, Nonthaburi, Samutsakorn and Nakhon Ratchasima. These provinces are situated nearby Bangkok, Thailand, in a radial distance of 250 km. 57 cabbage samples were purchased from local markets and 43 samples were from supermarkets. While 60 tomato samples were bought from local markets and 40 samples were from supermarkets. These samples were collected from October 2014 to October 2015. The samples were prepared in accordance to protocols proposed by Codex Alimentarius Commission (2015) prior to GC-MS/MS analysis for the pesticide residues. The representative portion (300–400 g) of the fresh vegetable was chopped into small pieces and blended using a food processor and mixed thoroughly. The homogenized samples were then extracted and analyzed as described below.

2.3 Sample preparation and GC-MS/MS analysis

The analysis of pesticide residues was conducted using the pesticide QuEChERS (Quick Easy Cheap Effective Rugged and Safe) method as explained previously (Anastassiades et al. 2003; Lehotay 2007; Lehotay et al. 2010). In concise, extraction of pesticides was executed by extracting 15 g of homogenized vegetable (cabbage or tomato) with 15 mL acetonitrile saturated with 6 g of magnesium sulfate and 1.5 g of sodium chloride. This extraction process was followed by a cleaning up procedure. For cabbage samples, this was performed by transferring the supernatant (1 mL) into another tube containing 50 mg of primary-secondary amine (PSA), 7.5 mg graphited carbon black (GCB) and 150 mg magnesium sulfate. For tomato samples, the similar procedure was applied but GCB was omitted from the cleaning up procedure. After agitating and centrifugation, the extracted supernatant was then transferred to an autosampler vial for direct injection into the Bruker GC-MS/MS system. The method validation in respect to recovery, reproducibility, calibration linear range, limit of detection (LOD), limit of quantification (LOQ) was carried out for the cabbage and tomato matrix as described previously (Dong et al. 2012; Koesukwiwat et al. 2011).

Detection of pesticides was accomplished using a Bruker 456 gas chromatography (GC) coupled with

Bruker Scion Triple Quadrupole mass spectrometer (GC-MS/MS). Details of GC-MS/MS conditions were detailed in the prior report (Duff and Voglino 2012). These pesticide analyses were done by multiple reaction monitoring (MRM) with 2 ion transition, at the experimentally optimized collision energy.

2.4 Calibration curves and quantification

Calibration curves of each pesticide were constructed and analyzed as detailed in the accepted methods (Dong et al. 2012; Koesukwiwat et al. 2011). This methodology was used each time when a new unknown sample set was analyzed. Aldrin was used as an internal standard. The peak area ratio of the pesticide standard to that of the internal standard was used for quantification. Excellent linearity and reproducibility of calibration curves were obtained as demonstrated by the coefficient of determination (r^2) values of >0.94 and relative standard deviations (RSD) of $<20\%$. The signal-to-noise ratio of pesticides of interest measured at 0.01 ppb was well above 30 for all pesticides of interest. Hence, detection limits were below 0.01 ppb using the sample preparation procedures portrayed above. Determination of pesticides in an unknown vegetable sample was performed in duplicate unless otherwise specified. Blank samples of cabbage and tomato used for recovery experiments and for preparation of the matrix-matched multi-level calibration standards, were checked and showed no pesticide residues. Recommended MRL values for each pesticide in the cabbage and tomato were referred to from MRL values launched by Thailand Ministry of Agriculture and Cooperation (2013), European Commission (2015), Codex Alimentarius Commission (2015), and Department of Food Safety, Japan (2006). The MRL values for each pesticide are provided in the tables of the Supplementary Data.

2.5 Statistical analysis

All data are expressed as mean $\pm$ standard deviation (SD). The mean differences of the parameters between two sample groups were assessed by either unpaired Student's t-test or the Mann-Whitney U-test, depending on their normality of distribution. p-values of less than 0.05 were deemed as statistically significant. SPSS version 18.0 (SPSS Inc. Chicago, IL, USA) was used for statistical analysis.

3 Results

3.1 GC-MS/MS method validation

Selection of 42 pesticides investigated was based on their common use in the cultivation of vegetables and fruits in Thailand. QuEChERS extraction method followed by GC-MS/MS analysis employed in this study, gave acceptable separation with good selectivity and high sensitivity for quantitative determination of the 42 pesticides of interest. Co-extracted interferences were not observed for all varieties of cabbages and tomatoes as revealed from blank extract analysis. There was no interfering peak co-eluted with analytes of interest as seen in the GC-MS/MS chromatograms of blank extracts from both cabbage and tomato samples (results are not shown). Also, in all samples of cabbage and tomato tested, there were no identifiable peaks detected with the same retention time as the internal standard, aldrin used in our GC-MS/MS assay. This verifies that using aldrin as the internal standard for our assay was applicable.

3.2 Types of pesticides found in cabbage samples

Of 42 pesticides, 10 were detected in the cabbage samples: carbaryl, carbofuran, λ -cyhalothrin, diazinon, dichlorvos, dimethoate, fenobucarb, isoprocarb, metalaxyl and pirimicarb (Fig. 1). Multiple pesticide residues were detected in all cabbage samples (100% positive). In general, fenobucarb and isoprocarb were the most frequently found pesticides in these cabbage samples (Fig. 1). The detection rate of isoprocarb in the cabbage samples was the highest at 100% (82/82). The second most detected pesticide in the cabbage samples was fenobucarb which was found in 74 of the 82 samples. This corresponds to a pesticide detection rate of 90%. The other 3 pesticides detected in the cabbage samples which were found at moderate rates were 63% (52/82) for dimethoate, 56% (46/82) for carbofuran and 52% (43/82) for λ -cyhalothrin. Metalaxyl, pirimicarb and dichlorvos were also found in the cabbage samples investigated although the occurrence rates were low <35% of the samples assayed for these pesticides. Carbaryl and diazinon were also detected at trace levels in the cabbage samples.

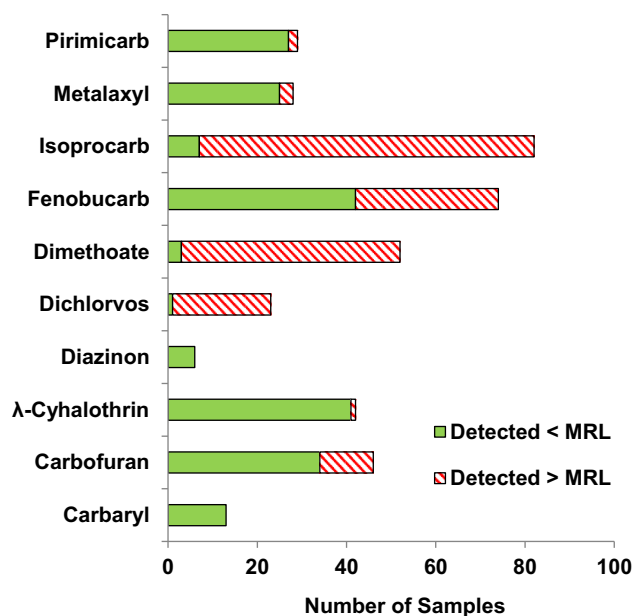


Fig. 1 Type of pesticides detected in the cabbage samples ($n = 82$) and detection frequency. The cabbage samples were purchased from local markets and supermarkets. For each pesticide detected, the samples were classified into two categories, samples detected with pesticide residue levels of < their MRL, and those detected with pesticide residue levels of > their MRL

3.3 Incidence of pesticide residues found in cabbage samples

Analysis of pesticide residues in 82 cabbage samples revealed that they were detected in all samples tested, representing a rate of pesticide detection rate of 100%. Out of the cabbage samples studied, 82 of them (100%) had pesticide residues exceeding the recommended MRL value. The main pesticide contributors that caused the MRL violence were isoprocarb, fenobucarb, dimethoate and dichlorvos (Fig. 1). This is because the levels of residues were found to exceed the MRL in most of the samples in which they were detected. For instance, of the 82 cabbage samples isoprocarb pesticide residues were detected in 91% (75/82) of the samples and had isoprocarb levels exceeding the MRL value (10 ppb). Similarly, 74 cabbage samples contained fenobucarb, 39% (32/82) of these samples contained fenobucarb exceeding the recommended MRL (300 ppb).

The incidence of pesticide residue detection in cabbage samples was compared among the samples collected from local markets and supermarkets in

Thailand. The results showed that all cabbage samples from either the local markets (57 samples) or supermarkets (25 samples) were contaminated with the investigated pesticide residues. This represents rates of pesticide residue detection of 100% regardless of where they were purchased from. Overall, the levels of pesticides found in all cabbage samples were above the recommended MRLs. This made the rate of MRL exceedance to be 100% in both the samples purchased from the local markets and those bought from the supermarkets.

3.4 Types of pesticides found in tomato samples

There were 11 different pesticides detected in the 100 tomato samples studied. These included carbofuran, chlorpyrifos, cypermethrin, diazinon, dichlorvos, ethion, fenobucarb, isoprocarb, λ -cyhalothrin, metalaxyl and pirimicarb (Fig. 2). A few tomato samples had only one pesticide while 97% had multiple pesticide residues. Among the monitored pesticides, fenobucarb, isoprocarb and cypermethrin were the most predominant pesticides found in the tomato samples. Of 100 tomato samples, fenobucarb was detected in 92 samples (this represents a 92% pesticide detection rate); isoprocarb was found in 85 samples (representing a 85% pesticide detection rate) and cypermethrin was detected in 72 samples (a 72% pesticide detection rate). Carbofuran, chlorpyrifos, λ -

cyhalothrin, dichlorvos, metalaxyl and pirimicarb were detected in the tomato samples at moderate rates (33–56%). While diazinon and ethion were also found in the tomato samples but with very low rates of detection (5 and 7%, respectively).

3.5 Incidence of pesticide residues found in tomato samples

Of the 100 tomato samples tested, pesticide residues were detected in all the samples, denoting a rate of pesticide detection of 100% in the tomato samples. In general, not all of the tomato samples had pesticide residues at levels that exceeded the MRL values. However, 82% of tomato samples contained pesticide residues with levels greater than the recommended MRLs. Interestingly, those samples with detected isoprocarb, most of them (82/85, Fig. 2) had levels of isoprocarb residues greater than the MRL value (10 ppb). The concentration range of isoprocarb residues found in the tomato samples was 1.9–2121 ppb. Only 5 samples in which fenobucarb was detected, had fenobucarb levels that exceeded the MRL (1000 ppb). The levels of fenobucarb residues detected in the tomato samples ranged from 0.1 to 3200 ppb. Levels of cypermethrin detected in the tomato samples ranged from 0.1 to 343 ppb. Of these, 4 samples had cypermethrin levels greater than its MRL (200 ppb). Only 3 tomato samples were detected with carbofuran levels greater than its MRL (100 ppb), and 4 samples were detected with ethion levels exceeding the recommended MRL (300 ppb, Fig. 2).

3.6 Comparison of incidence of pesticide residue detection in tomato samples from local markets and supermarkets

To differentiate the sources of produces, the data were examined according to whether tomato samples were purchased from local markets or supermarkets. The results showed that the profiles of pesticides detected in the tomato samples purchased from the supermarkets were similar to those bought from the local markets. 51 samples of the tomato samples bought from local markets (60 samples) contained pesticide residues, corresponding to rate of pesticide residues detection of 85%. Whereas, pesticide residues were detected in all the tomato samples purchased from the supermarkets (40 samples). This corresponds to the rate of pesticide residues detection of 100%. When the data was classified according to their MRLs, it was found that 52 tomato samples bought from local markets had

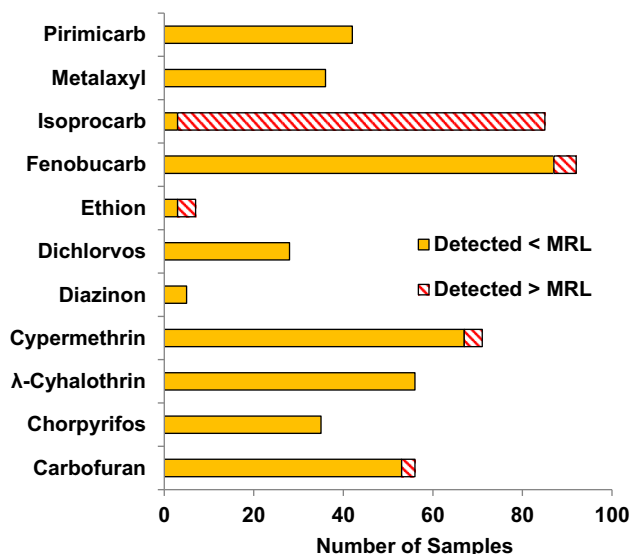


Fig. 2 Type of pesticides detected in the tomato samples ($n = 100$) and frequency of detection. The tomato samples were purchased from local markets and supermarkets. For each pesticide detected, the samples were classified into 2 groups, samples detected with pesticide residue levels of < their MRL, and those detected with pesticide residue levels of > their MRL

pesticide levels greater than the recommended MRL. This denotes that the rate of MRL exceedance was 87% for samples purchased from local markets. Whereas 29 tomato samples obtained from supermarkets contained pesticides at levels of greater than the MRL. This yields an occurrence rate of MRL exceedance of 73% in the samples bought from the supermarkets.

3.7 Effect of washing on pesticide removal from cabbage and tomato samples

The effect of washing with running water method on the removal of pesticide residues from the cabbage and tomato was investigated. Table 1 shows changes in pesticide residue concentration after washing cabbage samples with the running water method. Washing under running water significantly reduced ($p < 0.05$) the concentrations of carbofuran and dimethoate when compared with the control unwashed samples. After washing, the carbofuran residues were removed by 93%, and the dimethoate residues were removed by 65% (Table 1). For the other three pesticide residues, λ -cyhalothrin, fenobucarb and isoprocarb, the washing removed a considerable amount of the pesticide residues. However, the mean concentrations of these pesticides after washing were not significantly reduced ($p > 0.05$) as compared with those unwashed samples.

The effect of the running water washing method on pesticide residue removal from tomatoes shows Table 2. After washing, carbofuran and fenobucarb residues were eliminated by 58 and 40%, respectively, but these differences were not reach statistical significant ($p > 0.2$). For both cypermethrin and λ -cyhalothrin, residues were removed by 27% with running water washing method. However, with no significant difference ($p > 0.5$) between the mean

concentrations of these pesticides after washing and those unwashed samples (Table 2).

4 Discussion

The GC–MS/MS assays developed for analysis of pesticide residues in cabbage and tomato in our laboratory was further modified and based on the procedures used previously for Chinese kale (Wanwimolruk et al. 2015b). These methods included sample preparation using the QuEChERS procedure followed by GC-MS/MS analysis (Anastassiades et al. 2003; Lehotay 2007; Lehotay et al. 2010). These methods were proven to be reliable and accurate. They were shown to be appropriate and applicable for quantitative analysis of pesticide residues in two commonly consumed vegetables, cabbages and tomatoes. Results of assay validation included good recovery, sensitivity, selectivity, linear calibration curves, good reproducibility and accuracy supported the use of these methods.

Among the pesticides found in the studies vegetables, 8 pesticides were commonly detected in both cabbage and tomato samples. These were carbofuran, fenobucarb, isoprocarb, pirimicarb, diazinon, dichlorvos, λ -cyhalothrin and metalaxyl. Most of the detected pesticides are approved for usage in tomatoes or cabbages in Thailand. However, three pesticides, fenobucarb, isoprocarb and pirimicarb are being used in these crops without approval from the Thai Department of Agriculture. This information can be accessed from www.doa.go.th. The large number of pesticide residues detected in cabbages and tomatoes is not surprising as this has been also observed in the other 3 commonly consumed vegetables (chinese kale, pakchoi and morning glory) purchased in Thailand. This earlier study was carried

Table 1 Effect of running water washing on pesticide removal in cabbage samples ($n = 8$ each)

Pesticide	Log P^b	Pesticide concentration (ppb) ^a		% Removal
		Unwashed	Washed	
Carbofuran	1.8	135 $\pm$ 103	9 $\pm$ 7*	93
λ -Cyhalothrin	5.5	52 $\pm$ 41	23 $\pm$ 35	56
Dimethoate	0.7	13,300 $\pm$ 10,500	4700 $\pm$ 4900*	65
Fenobucarb	2.8	582 $\pm$ 632	246 $\pm$ 410	58
Isoprocarb	2.3	839 $\pm$ 781	405 $\pm$ 297	52

^a Pesticide concentration (ppb) in mean $\pm$ SD ($n = 8$)

^b Log P is partition coefficient (octanol/water) of a pesticide which was obtained from Pesticide Properties Database

* Statistically significant different from the control (unwashed) group ($p < 0.05$)

Table 2 Effect of washing (running water method) on pesticide removal in tomato samples (n = 8 in each group)

Pesticide	Log P ^b	Pesticide concentration (ppb) ^a		% Removal
		Unwashed	Washed	
Carbofuran	1.8	15 ± 33	6 ± 14	58
λ-Cyhalothrin	5.5	1.5 ± 1.4	1.1 ± 1.6	27
Cypermethrin	5.3	46 ± 120	33 ± 89	27
Fenobucarb	2.8	10 ± 7	6 ± 4	40

^a Pesticide concentrations (ppb) shown in mean ± SD (n = 8). No statistically significant differences between the control (unwashed) and the experimental groups (p > 0.05)

^b Log P is partition coefficient (octanol/water) of a pesticide which was obtained from Pesticide Properties Database

out by our research group in Thailand and 9 to 12 individual pesticides were detected in these 3 commonly consumed vegetables (Wanwimolruk et al. 2015b, 2016). The profile of pesticide residues detected in Chinese kale, pakchoi and morning glory were somewhat similar to our findings in cabbage and tomato samples. Most of the cabbage and tomato samples were found to contain more than one pesticide residue. In fact, 100% of cabbage and 97% of tomato samples tested had multiple pesticide residues. This suggests that Thai farmers tend to use a mixture of pesticides when growing cabbages and tomatoes. This is consistent with previous studies which found that most (80–90%) Chinese kale, pakchoi and morning glory bought from both local markets and supermarkets in Thailand, contained multiple pesticide residues (Wanwimolruk et al. 2015b, 2016). A mixture of pesticides is commonly used by Thai farmers because they believe this increases the protection of crops and to avoid resistance (Kaewboonchoo et al. 2015; Prasertsung 2016). The findings observed here also imply that pesticides are often and greatly used in the growing of these two commonly consumed vegetables in Thailand. Our observations are in agreement with earlier studies revealing that pesticides and fertilizers are utilized by most farmers in Thailand for cultivating vegetables (Kaewboonchoo et al. 2015; Ongkunaruk and Piyakarn 2011; Taneepanichskul et al. 2012).

Similarly, fenobucarb and isoprocarb were also found to be the most predominant pesticides in the cabbage and tomato samples investigated. Thus, it is apparent that these 2 pesticides, fenobucarb and isoprocarb are often used in growing cabbage and tomatoes in Thailand. In addition to these 2 pesticides, cypermethrin seems to be a popular choice of pesticide used in tomato cultivation. Although there were many reports on analytical methods for determination of fenobucarb and isoprocarb in vegetables and fruits (Koesukwiwat et al. 2011; Krause

1988; Li et al. 2015; Pihlstrom et al. 2007), little information exists on the detection of these two carbamate residues in cabbage and tomato. An early investigation conducted in Thailand (Vichapong et al. 2011) reported detection of 6 carbamates in a rather small sample of cabbage (n = 3). The carbamate pesticides found included isoprocarb, methiocarb, promecarb, carbaryl, carbofuran and propoxur. Interestingly, isoprocarb residues were detected in their cabbage samples at level of 220 ± 30 ppb (mean and SD). The mean isoprocarb level found in those cabbages was in the concentration range observed in our cabbage samples (2.4–3.590 ppb). Fenobucarb was reported to be found in tea leaves and some vegetables cultivated in Vietnam including mugwort, heartleaf, water spinach and star gooseberry (Pham et al. 2011). They also detected other pesticides including trichlorfon, cyfluthrin, and cypermethrin in vegetable samples. Their research showed that food products tested frequently met the upper limits for acceptable daily intake (ADI) levels. This demonstrated that existing pesticide management practices in Vietnam may pose risks to human health (Pham et al. 2011).

Of interest, second to fenobucarb and isoprocarb, the organophosphorus dimethoate was often found in the cabbage samples. The levels of dimethoate in the cabbage samples ranged from 14 to 191,877 ppb. Many cabbage samples (60%) contained dimethoate residues exceeded the recommended MRL (50 ppb). The highest one 191,877 ppb is equal to 192 mg/L. When the concentrations of dimethoate in the sprayed solution were calculated according to the dose recommended by the manufacturer. It appeared that the sprayed solutions could have dimethoate at concentrations of 300–400 mg/L. If these solutions are sprayed on the cabbages and the farmers do not follow the GAP guidelines, i.e. the cabbages are harvested a day after the pesticide was applied. This results in extremely high concentrations of

dimethoate on the cabbages, similar to those we detected in our cabbage samples. The detected concentrations could be even higher if the excessive doses of dimethoate are used. This MRL violation is likely to cause the exceedance of the acute reference dose (ARfD) and is of safety concern to consumers. However, as a risk assessment was not the primary objective of this study, this was not explored further. We are planning to conduct a risk assessment in the near future because we recognize its importance.

Overall, the present research found an incidence of pesticide detection in the cabbages and tomatoes investigated to be 100%, with occurrence rates of MRL exceedance of 100% for cabbage samples and 82% for tomato samples. These figures are very high and far greater than what would be found in developed countries. For instance, the American FDA (Food and Drug Administration) monitoring results showed that pesticide residues were detected in 30% of domestic and 21% of imported vegetables (Granby et al. 2008). This survey showed that 42% of samples contained pesticide residues less than the MRLs, but only 5% of the samples had pesticide residues above the MRLs (Granby et al. 2008). In Taiwan, 14% of vegetable samples were detected with pesticide residues (Chang et al. 2005). The monitoring program of pesticides carried out in the EU and Norway revealed evidence for low incidence of pesticide residues above the MRLs (Berger and Von Holst 2001).

On the other hand, in developing countries including China, Bangladesh, Pakistan and Thailand, there has been evidence showing that pesticide residues contamination exists in more than 50% of vegetables and fruits (Chowdhury et al. 2013; Chen et al. 2011; Phopin et al. 2017; Syed et al. 2014; Wanwimolruk et al. 2016). Residues of many organophosphorus pesticides such as chlorpyrifos, dimethoate, monocrotophos and profenofos were detected in 54% of the vegetable samples collected from India markets (Swarnam and Velmurugan 2013). A Thai study (Sapbamrer and Hongsisong 2014) found an overall rate of 25% pesticide detection in several market vegetables. When compared to the current study's findings of 100% pesticide detection in cabbages and tomatoes, the 2014 findings is considerably lower. The divergence is probably due to different kinds of vegetables, cultivation seasons, types of pesticides used and analytical assays employed. When studies conducted by our research group using the same analytical methods are considered, the data have shown similar trends of high incidence of pesticide detection in the vegetables sold in Thailand. The extremely high rates of

pesticide detection in 3 commonly consumed vegetables (Wanwimolruk et al. 2016), were consistent with the high rate of pesticide detection (i.e., 100%) in the cabbages and tomatoes seen in this study. These results not only indicate that many vegetables sold in markets of Thailand are cultivated using pesticides but also imply that pesticides are widely and extensively used in Thai agriculture.

For monitoring the incidence of pesticide detection and of MRL exceedance in Thailand, the analytical methods employed in the pesticide monitoring program are important. The FDA and the Ministry of Public Health of Thailand published information on the pesticide monitoring survey of fruits and vegetables including cabbages and tomatoes (Srithongkum 2014). The report was based on more than 60,000 samples per year. It revealed that violations (with pesticide concentration above their MRLs) observed in vegetables and fruits marketed in Thailand varied from 4% in 2013 to 5% in 2011. These figures released by the FDA of Thailand were much lower than the incidence of MRL exceedance found in the cabbage and tomato samples tested in the current investigation. The contradictory results are mainly due to the different methods used in the two screening surveys. Cholinesterase inhibition assay kit (GT-Test kit) was employed in the FDA of Thailand study. This particular test kit is only qualified for identifying two clusters of pesticides i.e. organophosphates (dicrotophos and EPN) and carbamates (carbofuran and methomyl). On the other hand, the existing GC-MS/MS method can identify several more clusters of pesticides (some 100–200 individual pesticides). This provides more sensitive, reliable and more accurate detection of pesticides than the GT-Test kit. Without using chromatographic technique, the kit assay detects only four individual pesticides, therefore it is less sensitive and does not provide an accurate finding of pesticide concentration (Wanwimolruk et al. 2015b).

In addition, other investigations conducted in developing countries have shown that some tomatoes, apples, grapes, melon, mango and plums contained pesticide residues above the MRLs recognized by the FAO/WHO. Wanwimolruk and coworkers reported that the MRL exceeded in 3 commonly consumed vegetables, ranging from 29 to 63%, which is exceptionally high compared to the rates found in developed countries (Wanwimolruk et al. 2015b, 2016). This high MRL exceedance is similar to that detected in our cabbage and tomato samples. This supports the findings of widespread pesticide use in agriculture in developing countries

such as Bangladesh, China, Pakistan and Thailand (Chen et al. 2011; Syed et al. 2014; Wanwimolruk et al. 2015b, 2016). As a developing country with agriculture as its main income, it is crucial for Thai farmers to use pesticides to protect crops from pests and ensure good harvests. The high rate of MRL exceedance seen in the cabbages and tomatoes indicates that Thai farmers do not use GAP with regard to pesticide doses. Also they do not allow enough time to elapse between pesticide application and the harvest of crops (standard pre-harvest intervals).

Interestingly, the unusual high MRL exceedance (100% for cabbage and 82% for tomato samples) was seen in the samples from both the supermarkets and local markets. The current study has also found that there was a resemblance in the pesticide profiles seen in the cabbages and tomatoes purchased from the 2 sources. This suggests that these vegetables are greatly contaminated with pesticide residues regardless of whether they are bought from the local markets or from supermarkets. Thai people have an expectation that supermarket product is of good quality regarding the levels of pesticide contamination and freshness. They are prepared to pay the higher prices for supermarket produce. The findings seen in this study with both cabbages and tomatoes bring up concern to the quality of the vegetables sold in supermarkets in Thailand. These findings also back up our previous findings (Wanwimolruk et al. 2016) which reported evidence of significant pesticide residues found in the three vegetables, i.e. chinese kale, pakchoi and morning glory sold in Thai supermarkets. Many produce sold in supermarkets have labels claiming to be pesticide-free or organic produce, yet these claims and labels are exhibited with no scientific proof or testing to verify them. The present findings put emphasis on the fact that these 2 commonly consumed vegetables that are sold in supermarkets in Thailand are not pesticide-free or organically cultivated. This contradicts with what the merchants claim on their produce labels. This equates to a food safety issue, as consumers' rights need to be protected. It is because the consumers buying produce labeled organic pay more and therefore should get better quality, pesticide free produce.

Since the national surveillance program of pesticide residues is not routinely enforced (Wanwimolruk et al. 2015a, b), the quality in regard to pesticide contamination of vegetables sold in Thailand is not assured. Our findings of considerable pesticide residues detected in both the cabbages and tomatoes purchased from supermarkets point to that

the contamination of pesticide residues in these frequently eaten vegetables cannot be guaranteed by the cost of the products or place of purchase. Because such high pesticide residues were found in vegetables collected from the markets in Thailand (Sapbamrer and Hongsisong 2014; Wanwimolruk et al. 2015b, 2016) this indicates that a substantial food safety issue exists in Thailand. This is an issue faced by the Thai government authorities including the Thai FDA and the Department of Agriculture. The funding sponsor of this study, the Agricultural Research Development Agency (Public Organization) of Thailand, hope for us to publicize our findings via the Thai government authorities. This will accelerate the implementation of effective policies and laws on pesticide residues as well as food safety. Relevant authorities of Thai government have been informed about our findings. The following actions have been intended; these include reinforcing the laws, properly establishing the GAP system to the farmers, continuing pesticide monitoring programs, and rectify the situation with the supermarket stakeholders. These are essential, not only to diminish the health risks to consumers linked with pesticide residues, but also to take care of consumers' rights.

Regarding the everyday methods to reduce pesticide contamination in vegetables, washing vegetables is perhaps the simplest and economic way to accomplish this. The present study has demonstrated that washing cabbage under running water significantly removed pesticide residues. The carbofuran was removed by 93%, and the dimethoate was removed by 65%. For other pesticide residues, λ -cyhalothrin, fenobucarb and isoprocarb, the percentage removal of the pesticide residues was less than those seen with carbofuran and dimethoate. The differences are likely to be related to their water solubility. Carbofuran and dimethoate have lower Log P (partition coefficient in octanol/water) values than λ -cyhalothrin, fenobucarb and isoprocarb (Table 1), thus carbofuran and dimethoate are being more water soluble compounds and more efficient to be removed by washing than the other three pesticides.

This finding is consistent with our earlier study in which washing chinese kale samples with water was found to be more effective to remove profenofos than cypermethrin (Wanwimolruk et al. 2015b). This is due to profenofos is more water soluble than cypermethrin. Similar to our own studies, other investigations have also indicated that there were differences in efficacy of removing individual pesticides from fruits and vegetables by washing. These disparities in

efficacy of washing with water may be linked with the pesticides' water solubility (Kong et al. 2012; Wanwimolruk et al. 2016). The results of the removal of pesticide residues from cabbage samples by washing are also in line with earlier studies in which it is shown that washing can diminish pesticide residues loosely attached to the surface of the vegetables (Abou-Arab 1999; Keikotlhaile et al. 2010).

Washing vegetables with water can remove the pesticide residues but not entirely. For example, by washing the tomatoes with water, chlorpyrifos residues were reduced by 41–44% (Rani et al. 2013). Washing asparagus with water decreased chlorpyrifos residues in asparagus by 24% (Chavarri et al. 2005). The profenofos residues in the Chinese kale were removed by 55% after washing with this method (Wanwimolruk et al. 2015b). This indicates that washing with water does not remove all pesticide residues from vegetables. The remaining pesticide residues found after washing were those pesticides have already been absorbed into plant components and cannot be washed away. The removal of pesticide residues from tomato samples by washing appeared to be less effective than those observed in the cabbage samples. The differences in the concentrations of pesticide residues between washed samples and unwashed samples were not significantly different (Table 2). Removal of pesticide residues by washing the tomatoes ranged from 27 to 58%. These results are consistent with those observed by Rani et al. (Rani et al. 2013). Washing tomatoes was reported to remove malathion residues by 47% (Reiler et al. 2015). Less efficient removal of pesticide residues from tomatoes by washing that was seen in our study may be explained by the distinct nature of tomato peels. A previous study (Clostre et al. 2014) has shown that the peels or rinds of vegetables including cucumber, pumpkin, yam and sweet potato, contained higher pesticide chlordecone residues than the edible part (pulp). This difference was accounted for by the composition of lipids and fibers (Clostre et al. 2014). Even though the composition of lipids and fibers of tomato peels was not determined, it is likely that tomato peels have greater lipid and fiber contents than those of cabbage leaves. Binding or deposition of pesticides to lipid and fiber compositions of tomato peel may hinder the removal of pesticide residues from tomatoes by washing.

Besides washing vegetables to remove pesticide residues before consuming, there may be other ways to eat cabbage with smaller amounts of pesticide residues being ingested. Fujita et al. found that there were differences in distribution of pesticide residues

in cabbage (Fujita et al. 2012). Majority of the pesticide residues (cypermethrin and acetamiprid) existed within the portions of the outer leaves of cabbage (approximately 80–97%), while relatively minimal residues existed in the portions of the core and middle leaves (1–9%). These results suggest that removal of the outer leaves of cabbage has an additional significant effect on removing the pesticide residues (Fujita et al. 2012). So eating the core and middle leaves of a cabbage will be safer than eating the whole cabbage because there are less pesticide residues ingested.

In spite of the fact that our findings uncovered possible risks associated with the intake of pesticide residues in these two commonly consumed vegetables, consumers may also be subjected to many of the same pesticides from a variety of foods. Overall, diet has to be brought into consideration to find out the true risks associated with exposure to pesticide residues. Pesticide residue exposure has been linked with increased risk of cancer and dysfunction of reproductive system in humans (Clostre et al. 2014; Ronn et al. 2011; Salerno et al. 2014; Uysal et al. 2013). Fruits and vegetables containing pesticide residues are greatly improved through exposing the contaminated fruit or vegetable to high heat by blanching, boiling and frying, lessening the pesticide intake into the body (Granby et al. 2008). This was demonstrated in pepper and eggplant where blanching and frying has been revealed to lead to a very large decrease (almost 100%) in the deposit of pesticide (Radwan et al. 2005). Nevertheless, vegetables like cabbage and tomato are often eaten raw (fresh) in some dishes. Thus, prior to consumption, if these vegetables contain a lot of pesticide residues and are not properly washed, the amount of pesticide ingested will be considerably large. The occurrence of pesticide detection exceeding the MRL in the cabbages and tomatoes observed in this work was surprisingly high suggesting there are potential health risks to Thai consumers. It should be noted that MRLs represent the limit of products, not the limit of safety (MacLachlan and Hamilton 2010). For that reason, if an MRL is exceeded, one cannot guarantee the safety of consuming such produce, i.e. fruits and vegetables. Although, whether there is a real issue or not, it has to perform more investigations by running a precise assessment of consumption risks. In this study, nevertheless, the data obtained were not used to assess the health risks linked with consuming cabbages and tomatoes. We expect to conduct this in a future investigation with a greater sample size.

5 Conclusion

In summary, this study of pesticide residues in cabbages and tomatoes clearly supports the previous findings that pesticides are widespread and extensively used in Thai agriculture. The results also confirm that contamination of pesticide residues in vegetables sold in Thailand is a major health issue. This challenges the Thai government authorities to pay more attention to resolve this national problem. Based on these findings, the National Pesticide Monitoring Program in Thailand should be enforced more effectively. This would also assist in the risk assessment of consumers' exposure to the harmful pesticide residues. However, in most developing countries like Thailand, resources including modern equipment (GC-MS and LC-MS) and qualified laboratory personnel are very limited. The quality and the pesticide scope of the laboratories need to be evaluated to ensure high standards and reliable results. Quality assurance and standardization of analytical methods among laboratories are equally important. In Thailand, it has been suggested to establish a pesticide residue laboratory-network that will help to reduce costs for the National Pesticide Monitoring Program. Also this will be fruitful for knowledge exchange.

To reduce the health risks from pesticide contamination, Thai authorities need to raise understanding and awareness among Thai farmers about the dangers of pesticide contamination. It is essential to educate farmers to understand the safe and proper use of pesticides following GAP standards. Greater resources for the surveillance and enforcement of pesticide levels are also desirable to warrant public safety from consuming contaminated fruits and vegetables. Educating consumers about properly washing vegetables prior to consumption is also recommended. This can reduce the amount of pesticide in our everyday intake.

Our conclusions are based on the evidence of pesticide contamination found in cabbages and tomatoes that are sold domestically. This excludes the fresh fruits and vegetables grown for export, as these are generally cultivated and harvested to GAP standards. The fruits and vegetables intended to be exported are compulsorily monitored for pesticide residues before exportation. Thus, this produce is guaranteed to be of good quality and of low incidence of pesticide contamination.

In Asia, although many countries have not established an effective National Pesticide Monitoring

Program but their government authorities are aware of pesticide contamination in food commodities and their toxicity. This report is the 5th research article in our series on food safety in Thailand. It is hoped that our studies will further stimulate alertness and understanding of this food safety problem, especially in Thailand and nearby Asian countries.

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Compliance with ethical standards

Conflict of interest The authors declare they have no conflict of interest.

Availability of data and materials Data can be found in Supplementary files.

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